



Daipayan Mandal

Assistant Professor | Civil Engineering | Structural Engineering

Kavikulguru Institute of Technology and Science (KITS),
Ramtek, Maharashtra, India

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GitHub: github.com/mdaipayan



PROFESSIONAL PROFILE

Assistant Professor and structural engineering professional with 12 years of academic service experience in Civil Engineering. Specialized interests include computational structural analysis, matrix methods, the Direct Stiffness Method (DSM), structural optimization, and engineering software development. Demonstrated ability to combine engineering theory, programming, research, and academic administration to develop practical technology-enabled solutions for structural analysis, engineering education, and examination process automation.

CORE SKILL SET

Structural Engineering	Structural Analysis, Matrix Methods, Direct Stiffness Method, Steel Structures, Truss Analysis, Structural Optimization
Computational Engineering	Numerical Methods, Engineering Computation, Structural Solvers, Algorithm Development, Engineering Visualization
Software Development	Python, Streamlit, Git, GitHub, Web-Based Engineering Applications, Computational Prototyping
Research	Structural Optimization, Computational Structural Mechanics, Engineering Education Technology, Academic Process Automation
Academic Administration	Examination Tabulation, Result Processing, Grading Workflows, Academic Documentation, Student Mentoring
Technical Documentation	LaTeX, Technical Reports, Academic Writing, Engineering Documentation

EDUCATION

Ph.D. Scholar in Applied Mechanics	<i>Ongoing</i>
Visvesvaraya National Institute of Technology (VNIT), Nagpur	
<i>Milestone:</i> Cleared the Ph.D. Entrance Test (VNIT).	
M.Tech. in Structural Engineering	<i>2015–2017</i>
Rashtrasant Tukadoji Maharaj Nagpur University (RTMNU), Nagpur	
B.E. in Civil Engineering	<i>2006–2010</i>
Kavikulguru Institute of Technology and Science (KITS), Ramtek	

PROFESSIONAL EXPERIENCE

Assistant Professor – Department of Civil Engineering	<i>Jan. 2013 – Present</i>
Kavikulguru Institute of Technology and Science (KITS), Ramtek, Maharashtra	

Simplex Infra-Structure Pvt. Limited

- Teach Civil and Structural Engineering subjects with emphasis on structural analysis, matrix methods, and the Direct Stiffness Method.
- Mentor undergraduate students in structural engineering, computational methods, engineering projects, and technical problem solving.
- Evaluate End Semester Examination answer scripts and contribute to academic and examination-related activities.
- Serve as Tabulation Incharge and contribute to examination result processing, grading workflows, and academic documentation.
- Developed an examination tabulation and document-generation system using Google Sheets as the primary data platform, without relying on commercial or proprietary ERP software.
- Automated the generation of examination admit cards and grade sheets using LaTeX-based document templates and structured data from Google Sheets.
- Introduce programming and computational tools into engineering teaching and project-based learning.

TEACHING AREAS

- Structural Analysis and Matrix Methods
- Strength of Materials and Solid Mechanics
- Steel Structures and Structural Design
- Computational Structural Engineering
- Engineering Optimization
- Civil Engineering Computing and Programming

E-CONTENT & TEACHING RESOURCES

Strength of Materials – E-Content Portal

Developed and curated web-based teaching resources for Strength of Materials, providing students with structured digital learning content and accessible reference material for engineering study.

Open E-Content / Teaching Resources



E-Content

RESEARCH INTERESTS

- Computational Structural Mechanics
- Direct Stiffness Method and Matrix Structural Analysis
- Structural and Truss Optimization
- Artificial Intelligence and Optimization in Structural Engineering
- Engineering Software and Web-Based Structural Tools
- Technology-Enabled Engineering Education

SELECTED PUBLICATIONS

Scan the QR code to access the publication or preprint.

Mandal, D. (2026). *Development of an Interactive Web-Based Tool for 2D Truss Analysis Using the Direct Stiffness Method.*

Computer Applications in Engineering Education, 34(3), e70183.

DOI: 10.1002/cae.70183



Mandal, D. (2025). *Automating Examination Grading and Tabulation: A Hybrid Policy-Compliant Framework for Higher Education.*

Preprint, TechRxiv.

DOI: 10.36227/techrxiv.175977600.03855876/v1



Mandal, D. (2019). *Compressive Strength Behavior Glass Fibre Reinforced Blast Furnace Slag-Based Material.*

SSRN Electronic Journal.

DOI: 10.2139/ssrn.3380074



OTHER ACADEMIC OUTPUT

Authored Book

Mandal, D., & Yadav, R. S. (2026). *Engineering a Greener Future: Principles of Environmental Science and Sustainability.* Notion Press.

Amazon book listing

Scan the QR code to view the book on Amazon.



Selected Conference Papers

- Mandal, D. (2021). *Compressive and Flexural Behaviour of Glass Fibre Reinforced Blast Furnace Slag Based Material.* Sustainable Environment and Infrastructure. ISBN: 978-3-030-51354-2.
- Mandal, D. (2020). *Compressive Strength Behaviour of Glass Fibre Reinforced Blast Furnace Slag-Based Material.* 11th International Conference on Geosynthetics (11ICG).
- Mandal, D. (2017). *Compressive Strength Behaviour Glass Fibre Reinforced Blast Furnace Slag Based Material.* Indian Geotechnical Conference 2017 (GeoNEst).

ENGINEERING SOFTWARE & DIGITAL PROJECTS

AI Truss Optimization Application

Developed an interactive Streamlit-based web application for truss optimization, integrating computational optimization techniques with an engineering-focused user interface.

Live Application

Automated Purlin Design Prototype (Ongoing)

Developing a Python/Streamlit application for automated purlin calculations, engineering checks, and structural design report generation.

Live Prototype



Web-Based Truss Solver

Developed a structural analysis solver implementing the Direct Stiffness Method. Deployed as a web-based engineering application with GitHub-based development and CI/CD workflow.

Live Application



3D Frame Analyzer & Beam Analysis Tools

- Developed computational engineering tools in Python and Streamlit for structural analysis, engineering calculations, and visualization.

INDEPENDENT SOFTWARE & COMMERCIAL PROJECTS

WardMentor – Education Technology Platform

Developed an independent web-based education technology platform demonstrating application of software development, database-driven workflows, and academic / administrative process digitization.

Live Demo



ShalaOS – School Management Platform

Developed an independent school-management platform focused on digitizing academic and administrative workflows for educational institutions.

Live Demo



ACADEMIC ADMINISTRATION & INSTITUTIONAL CONTRIBUTIONS

- **Institutional Duty – Tabulation Incharge, Examination and Evaluation Cell:** Responsible for examination tabulation, result processing, coordination of evaluation-related records, verification of grading data, and preparation of examination documentation.
- Developed an examination tabulation system using Google Sheets as the primary data platform, replacing the need for a commercial/proprietary ERP for the tabulation workflow.
- Implemented LaTeX-based automated generation of examination admit cards and grade sheets from structured tabulation data.
- Contribute to academic grading and result-verification procedures associated with End Semester Examinations.
- Evaluate End Semester Examination answer scripts and support examination-related academic activities.
- Student Mentoring and Academic Counselling, including support for technology-assisted academic processes.

PROFESSIONAL ACTIVITIES

Extension Activity: Active participant in the Rural Happiness Program in Ramtek.

DIGITAL PROFILE

ORCID orcid.org/0000-0002-5096-3417

Research / Publications See QR codes beside selected publications for direct access

Engineering Applications See QR codes beside selected live software projects

DECLARATION

I hereby declare that the information provided in this curriculum vitae is true and correct to the best of my knowledge.

Daipayan Mandal
Assistant Professor, Civil
Engineering
KITS, Ramtek